MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLIES, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the general requirements for circuit card assemblies, hereinafter referred to as "assemblies", which form part of the Electronics System Test Set type AN/TSM-100.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-f-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Fquipment and Tools for Ordnance Material: Packaging of
MIL-I-45607	inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection of
MIL-T-60876	Test Set, Servo Amplifier, TS-2658/TSM-100
MIL-T-60877	Test Set, Distribution Box, TS-2659/TSM-100
MIL-T-60878	Test Set, Sight Current Generator TS-2660/TSM-100

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MIL-T-60879	Test Set, Power Panel, TS-2661/TSM-100

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods.

(Copies of specifications, standards, drawings and packaging data sheets, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

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3. REQUIREMENTS

3.1 Fabrication. The assemblies shall be manufactured in accordance with the applicable drawings referenced in the detail specification.

3.1.1 Detail specifications. Detail requirements of assemblies shall be as specified in the applicable detail specification identified in Table I.

3.2 General Specifications. Where applicable, the assemblies shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence.

b. Dimensions and tolerances.

c. Inorganic protective surface finishes.

d. Workmanship.

3.2.2 MIL-C-46357.

a. Etched circuit assembly. The requirements for etched circuit assemblies of MIL-C-46357 shall apply with modification as follows:

1. Plated etched circuit boards. In addition, assemblies shall not contain copper or nickel visible through hot solder dip, when viewed at 10 times magnification.
2. Location grid. The component mounting holes shall be within 0.007 inch of the true grid intersection, instead of 0.005 inch.
3. Transistor and diode leads. Transistor and metal cased diode leads shall be covered with sleeving in areas where there is a possibility of shorting. No color coding of sleeving shall be required.
4. Drying. The drying process shall not be required.

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Table I. Detail specification identification

Specification MIL-C-60898	Title	Assembly Part No.	Application	
			Test Set	Specification
/1	Circuit Card Assembly, Logic Function Resistors	10554163	TS-2659/ TSM-100	MIL-T-60877
/2	Circuit Card Assembly, Relay Sense Resistors	10554184	TS-2659/ TSM-100	MIL-T-60877
/3	Circuit Card Assembly, Relay Sense and K12 Operation	10554179	TS-2659/ TSM-100	MIL-T-60877
/4	Circuit Card Assembly, K6 Operation and Sequence	10554185	TS-2659/ TSM-100	MIL-T-60877
/5	Circuit Card Assembly, K8 Operation and Sequence	10554164	TS-2659/ TSM-100	MIL-T-60877
/6	Circuit Card Assembly, K11 Operation and Sequence	10554165	TS-2659/ TSM-100	MIL-T-60877
/7	Circuit Card Assembly, Relay Signal Inverters	10554189	TS-2659/ TSM-100	MIL-T-60877
/8	Circuit Card Assembly, Noise Suppression	10554208	TS-2659/ TSM-100	MIL-T-60877
/9	Circuit Card Assembly, Sight Current Generator	10554188	TS-2660/ TSM-100	MIL-T-60878
/10	Circuit Card Assembly, Power Unit	10554186	TS-2661/ TSM-100	MIL-T-60879
/11	Circuit Card Assembly, Servo Amplifier Test Input	10554194	TS-2658/ TSM-100	MIL-T-60876

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3.3 Environmental service conditions

3.3.1 Vibration (non-operating). The assemblies shall be subjected to a simple harmonic motion having excursion as follows:

Frequency (Hz ± 2 percent	Excursion (double amplitude) (inches ± 10 percent)
5 to 15	0.06
15 to 25	0.04
25 to 55	0.02

The frequency shall be varied logarithmically between 5 and 55 cycles per second (cps) within 1.5 minutes and returned to 5 cps in 15 minutes. The motion shall be applied to the assemblies for a period of 30 minutes in each direction of the 3 mutually perpendicular axes (total of 90 minutes), relative to its normal operational position. Subsequent to vibration, the assemblies shall not be damaged or adversely affected and shall meet the electrical requirements specified in the detail specification.

3.3.2 Temperatures.

3.3.2.1 Operating. The assemblies shall meet the requirements of the detail specification while exposed to temperatures of + 32°F and +131°F.

3.3.2.2 Storage. The assemblies shll meet the requirements of the detail specification at a temperature of 60°F and 90°F after having been exposed to each temperature of -80°F and +185°F.

3.3.3 Hymidity: The assemblies shall function as specified in the detail specification and shall not suffer any damage or deterioration which shall impair satisfactory performance after being subjected to ten consecutive humidity cycles at. relative humidities between 85 and 95 percent and temperatures from 82 °F to 160°F over a period not less than 240 hours.

3.3.4 Shock. The test set shall show no evidence of damage or be adversely affected after being subjected to a total of 18 shocks consisting of three shocks in opposing directions along each of the three mutually perpendicular axes. Each shock shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11±1 millisecond.

3.3.5 Altitude.

3.3.5.1 Operating. The test set shall not be damaged or adversely

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affected when exposed for a period of not less than four hours to conditions encountered at an altitude of 10,000 feet above sea level. During and subsequent to such exposure the test set shall meet the requirements of 3.4.

3.3.5,2 Non-Operating. The test set shall meet the requirements of 3.4 after exposure to conditions encountered at 50,000 feet above sea level for a period of not less than one hour. The test set shall not be damaged or be adversely affected due to such exposure.

3.3.6 Sand and dust. The equipment shall withstand, in both an operating and nonoperating condition, exposure to sand and dust particles as encountered in desert areas.

3.3.7 Salt spray. Equipment shall withstand, in both an operating and nonoperating condition, exposure to salt-sea atmosphere.

3.3.8 Fungus-inert materials. The requirements for fungus-inert materials shall be in accordance with MIL-STD-454, Requirement 4.

3.4 Electrical Performance. Electrical performance of assemblies shall be as specified in the detail specification.

3.4.1 Wire (hookup). Stranded conductors of suitable degrees of flexibility s all be used whenever practicable within the require-
‘merits of this specification. Hookup wire shall be of such cross section and temper as to provide ample and safe current-carrying capacity and mechanical strength. In general, wire shall not be smaller than AWG No. 22. Wire directly associated with the primary power circuit shall be of a size consistent with the current requirements of the circuit. When it can be demonstrated that the use of smaller hookup wire will result in no “loss of performance and at the same time will produce benefits, such as reduction in weight or improved accessibility, smaller wire may be used. This provision is made specifically to permit the use of smaller wire in cables having a large number of wires and having adequate support against adverse effects of vibration. The size of wire leads supplied with parts shall be controlled by the specification for these parts.

3.4.1.1 Insulation on conductors. Insulation of the sleeve type *” may be used, provided its length is sufficient to prevent slippage along the conductor to such an extent as to expose a section of the conductor that should be insulated. Insulation shall provide adequate dielectric strength and adequate leakage resistance when the equipment is operated under the designated service conditions. The mechanical and electrical properties of the insulating materials and impregnating compounds shall not be permanently impaired

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as a result of being exposed to extremes of temperature incurred in the equipment. Rubber without braid may be used only when the insulation is not depended upon to space or support the conductor, or when a number of conductors are assembled in the forms of cordage and the assembly provided with a braid covering which has been suitably varnished or lacquered. All conductor insulation shall be of the noncombustible or slow-burning type; that is, with the insulated conductor held in a horizontal position in still air, self-sustained combustion of the insulation or lacquer shall not progress at a rate in excess of 1 inch in 1 minute.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4,2 First article (initial production) approval. The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests snail be 3 assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used-in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsibel for co ducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's Inspection.

4.3 Inspection provisions.

4,3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

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4.3.2 Examinations and tests. Unless otherwise specified, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests to verify requirements of 3.4 shall be performed in accordance with test methods arid procedures in the detail specification. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.3.2.1 Control tests. One assembly shall be selected at random as a control sample from each 100 produced. The control samples shall meet the requirements and tests of Table II after they have met the requirements and tests of 4.3.4.

Table II

Test	Requirement	Test Procedures
Fabrication	3.1	Applicable drawings - visual
General specifications	3.2	Visual-tactile
Vibration	3.3.1	4.5.1.1, 4.5.1.2
Temperature	3.3.2	4.5.1.1, 4.5.1.4
Humidity	3.3.3	4.5.1.1, 4.5.1.3
Shock	3.3.4	4.5.1.1, 4.5.1.5
Altitude	3.3.5	4.5.1.1, 4.5.1.6
Sand and dust	3.3.6	4.5.1.1, 4.5.1.7
Salt spray	3.3.7	4.5.1.1, 4.5.1.8
Fungus	3.3.8	4.5.1.1, 4.5.1.9

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal Or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Functional tests. Detail functional requirements specified in the detail specification shall be inspected on a 100 percent basis, unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50°F to +90°F
b.	Relative humidity	90% RH max.
c*	Pressure	28 to 32 inches of Hg

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4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design. The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in the detail specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926, Since Tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerances limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment. Test equipment used to test assemblies for verification of Section 3 electrical requirements shall be as specified in the detail specification.

4.5 Test methods and procedures.

4.5.1 Environmental,

4.5.1.1 Test conditions. The assemblies shall be mounted in an upright position in a manner similar to that in which it will be in service and shall be subjected to the environmental tests of the following subparagraphs, The atmospheric pressure shall be room ambient. Except for the high and low temperature tests or the humidity tests, as applicable, the temperature and humidity shall be room ambient.

4.5.1.2 Vibration. The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.10 The measuring instruments of the testing device shall be capable of measuring the vibration amplitude-within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements

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specified in 3.3.1. Subsequent to the vibration test, the assembly shall show no damage or defect resulting from the test when examined by visual and tactile means. The assembly shall then be inspected to determine compliance with requirements specified under 3.4.

4.5.1.3 Humidity. The humidity test shall be performed in accordance with Procedure J of Method 507 in M1L-STD-810. Subsequent to the humidity test, the assembly shall be tested to determine compliance with 3.3.3.

4.5.1,4 Temperature. Test in accordance with Procedure I of Method 501 in MIL-STD-810 for high temperature operating and non-operating under conditions specified in 3.3.2. Test in accordance with Procedure I of Method 502 for low-temperature operating and non-operating under conditions specified in 3.3.2.

4.501.5 Shock. The shock test shall be performed in accordance with MIL-T-5422.

4.5.1.6 Altitude. The altitude test shall be performed in accorc ante with the following steps.

Step 1. Place the assembly in the test chamber in a manner that will simulate service usage, making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in sevice shall be adequately covered. For tests where temperature values are controlled the test chamber shall be at standard ambient conditions when the assembly is installed. The assembly shall then be operated to determine that no malfunction or damage was cause due to faulty installation or handling. The requirement for operation following installation of the assembly in the test facility is applicable only where operation is required during or immediately following exposure to the specified test. Standard ambient temperature shall be maintained during the entire test.

Step 2. Reduce the chamber internal pressure to 87.5 mm of H (3.44 in. of Hg or 50,000ft. above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

Step 3. Increase the chamber pressure to 523 mm of Hg (20.6 in. of Hg or 10,000 ft. above sea level) and then operate the assembly and compare the results with the data obtained in

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accordance with pretest performance record made of all data necessary to determine compliance with required performance.
Step 4. With the assembly not operating, return the chamber to standard ambient conditions.
Step 5. Operate the assembly and compare the data obtained with the data obtained in accordance with pretest performance record.
Step 6. The assembly shall be visually inspected and a record made of any damage or deterioration resulting from the test. If a test chamber is used for the test, perform a visual inspection of the test within the chamber at test conditions, when possible. Upon completion of the test, visually inspect the assembly again after the assembly has been returned to standard ambient conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external parts which in any manner prevent the assembly from meeting operational *service* or maintenance requirements shall provide reason to consider the assembly as having failed to withstand the conditions of the test.

4.5.1.7 Sand and dust. The sand and dust, test shall be performed in accordance with MIL-T-5422.

4.5,1.8 Salt spray. The salt spray test shall be performed in accordance with MIL-T-5422,

4.5.1.9 Fungus. The fungus resistance test shall be performed in accordance with MIL-T-5422.

4.5.2 Wire insulation. Wire insulation requirements per 3.4.1 shall be verified by certification.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation,. packaging, packing, and marking shall be in accordance with the Packaging Data Sheet specified in the detail specification,

6. NOTES

6.1 Intended use. The circuit card assemblies covered by this specification are plug-in type components intended for use in test equipment as listed in Table I.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number and date of the applicable detail specification (see Table I).

c. Circuit card assembly part number (see Table I).

(1. Instructions for arranging of examination, test, and approval of the first article (see 4.2).

Custodian:

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Preparing Activity:

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Project No. 4931-A039

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced documents(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

YES NO (If "yes" in what way?)

4 REMARKS (Give any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

5 SUBMITTED BY (Printed or typed name and activity - Optional) DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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